

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MRBENCH | Context: Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is a LOWER-priority dimension. MRBench is text-based English dialogue [Q13, Q133], encoded as conversation history with tutor/student turns and lengths measured in characters [Q147]. The deployment is also text-only English-medium via mobile or enterprise software and explicitly targets fluent English speakers; no modality, script, or language mismatch exists. Mobile internet penetration in metropolitan India is high (75%+ of children aged 5–14 use mobile phones, with metro rates higher) [WEB-10, WEB-12], and Android dominates with 92.4% market share [WEB-13], so text-based delivery on mobile is well-supported. The minor concerns are that conversation lengths and turn structures differ between Bridge and MathDial [Q141, Q142] — but this represents internal diversity rather than a target-context mismatch.

ID	Evidence
Q13	“Formally, let’s define the conversation history between a tutor and a student as $H = \{(T1, S1), (T2, S2), \dots, (Tt, St)\}$, where T_i represents the i -th response from the tutor, and S_i represents the i -th response from the student. Let S_k denote the student’s most recent k utterances, where $k \in [1, \dots, t]$, containing a mistake or confusion. Then the objective of the tutor is to provide the most appropriate response T_{t+1} to address this mistake or confusion.” (p.3)
Q133	“The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2.” (p.12)
Q147	“In all cases, length is estimated using the number of characters.” (p.14)
Q141	“It can be observed that the conversation history and response lengths from different LLMs and humans are generally shorter in the Bridge dataset compared to the MathDial dataset.” (p.14)
Q142	“Additionally, the number of turns differs between them.” (p.14)
WEB-10	Mobile internet broadly available in Indian metros for text delivery (https://datareportal.com/reports/digital-2024-india)
WEB-12	~65% of Indian children aged 5–14 used mobile phones in 2023 (https://www.dataforindia.com/comm-tech/)
WEB-13	Android dominates Indian smartphone OS market at 92.4% in 2025 (https://www.imarccgroup.com/india-smartphone-market)

Strengths

- Text-only English format perfectly matches the deployment’s text-based English-medium delivery [Q13]
- Conversation-history input format directly matches a mobile-app tutoring dialogue use case [Q13, Q133]
- Internal diversity in conversation length across Bridge and MathDial provides some signal-distribution variation [Q141, Q142]

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Q133	“The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2.” (p.12)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are aligned: both benchmark and deployment are English-medium text dialogues with conversation history and turn structure [Q13, Q133]. No script or language mismatch.

IF-2: Regional infrastructure supports the format. India has approximately 750M+ mobile internet users with strong metropolitan 5G coverage [WEB-10, WEB-11], and Android dominates at 92.4% market share [WEB-13].

IF-3: Domain-specific form is well-matched: tutor/student dialogue turns over text are the deployment's actual use case. No multimodal (audio, image, handwriting) requirement is specified.

IF-4: Minor gap: MathDial does not provide conversation topic labels [Q145], which could complicate routing/curriculum-tagging in deployment. Otherwise no form mismatch identified.

ID	Evidence
Q13	"Formally, let's define the conversation history between a tutor and a student as $H = \{(T1, S1), (T2, S2), \dots, (Tt, St)\}$, where Ti represents the i -th response from the tutor, and Si represents the i -th response from the student. Let Sk denote the student's most recent k utterances, where $k \in [1, \dots, t]$, containing a mistake or confusion. Then the objective of the tutor is to provide the most appropriate response T_{t+1} to address this mistake or confusion." (p.3)
Q133	"The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2." (p.12)
WEB-10	Mobile internet broadly available in Indian metros for text delivery (https://datareportal.com/reports/digital-2024-india)
WEB-11	Mobile broadband dominates Indian internet access; 5G rollout in metros since 2022 (https://muftinternet.com/blog/usage-statistics-internet-and-mobile-users-in-india-2025/)
WEB-13	Android dominates Indian smartphone OS market at 92.4% in 2025 (https://www.imarcgroup.com/india-smartphone-market)
Q145	"The conversation topic is not provided in MathDial." (p.14)

Information Gaps

- No data on actual deployment device distribution for Grade 1–8 metropolitan students specifically (smartphone vs. tablet vs. laptop share)

Requires Expert Verification

- Whether deployment intends to support voice/audio modalities later, which would require benchmark extension